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The effect of corporate governance mechanisms on European Mergers and Acquisitions

Ioannis Tampakoudis, Michail Nerantzidis, Demetres Soubeniotis and Apostolos Soutsas

Abstract

Purpose – The purpose of this study is twofold: First, to assess the economic impact of Mergers and Acquisitions (M&As) on European acquiring firms from the beginning of the sixth merger wave onward. And second, to investigate the effect of CG mechanisms such as board size, voting rights and anti-takeover provisions (ATPs) on acquirers' gains, along with a set of control variables.

Design/methodology/approach – For the purpose of the study, the authors use a sample of 349 completed M&As across all business sectors between European firms from 01/01/2003 to 31/12/2017. Abnormal returns are estimated by applying an event study methodology, and the effects of CG mechanisms are assessed with univariate and multivariate cross-sectional regressions.

Findings – The authors present evidence that acquirers realize significant positive excess returns upon the announcement of M&As. The authors find past profitability to be a strong indicator of value creation, while most of the traditional firm-specific and deal variables fail to interpret the results. The authors' analysis indicates that the examined CG measures have a significant effect on acquirer's gains. More specifically, the authors find that boards in excess of eight directors are negatively related to announcement-period abnormal returns. In contrast, the wealth effects for acquiring firms are positively related to shareholders' voting rights and/or to the number of ATPs. The estimated coefficients of all three CG mechanisms are statistically significant across alternative model specifications.

Research limitations/implications – A clear implication is that the existence of certain CG mechanisms leads to value-enhancing strategic decisions for European acquirers. In terms of policy direction, the authors' findings assist practitioners and/or national and transnational institutions in perceiving the efficacy of certain CG practices.

Practical implications – This study indicates that Corporate Governance Statements (CGSs) fail to provide adequate information to investors to understand in-depth the CG mechanisms that companies apply. Thus, the authors recommend that CGSs should provide not only narrative information but also information that may generate value for shareholders and other stakeholders as well. Such information should be qualitative and/or quantitative in nature and be made available to market participants to support their decision-making.

Originality/value – To the authors knowledge, this is the first study that investigates the effect of CG on the economic impact of M&As for European acquirers, using three widely examined CG mechanisms, namely, the board size, the voting rights and the ATPs. The authors' empirical findings form the basis for further examination of the linkage between M&As and CG, with the intention of establishing the appropriate CG framework that will ensure shareholder wealth creation. This line of research could produce new insights in the field, allowing investors and policymakers to appreciate the benefits of effective CG.

Keywords Europe, Corporate governance, Event study, Acquirer abnormal returns, Mergers and acquisitions

Paper type Research paper

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1. Introduction

The main drivers that force firms to engage in Mergers and Acquisitions (M&As) are the creation of synergies, the assimilation of new competencies and the diversification of

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business risk, which in turn strengthen corporate growth and enhance corporate profitability (Alexandridis *et al.*, 2010). This illustrates that M&As may act as an efficient way for firms to expand into new markets and increase the wealth of their shareholders (Papadakis, 2005). In that context, during the past decades academic and market research has extensively studied the wealth effects deriving from the activity of M&As (Alexandridis *et al.*, 2010; Alexandridis *et al.*, 2017; Masulis *et al.*, 2007). One of the main findings of the literature is that M&As often enough fail to generate value for the shareholders of acquiring firms. Thus, one of the purposes of this study is to examine the economic impact of M&As on European acquirers.

Considering that in practice managers do not always take value-enhancing strategic decisions (Masulis *et al.*, 2007), a growing body of research focuses on CG mechanisms as a means for effective decision-making in terms of shareholder value creation (Alexandridis *et al.*, 2010; Bebchuk and Cohen, 2005; Bebchuk *et al.*, 2009; Cremers and Nair, 2005; Gompers *et al.*, 2003; Masulis *et al.*, 2007). For this reason, the second purpose of our study is to investigate the effect of CG mechanisms on acquirers' gains. In particular, there is a considerable number of corporate mechanisms – both internal and external – that induce the self-interested controllers of a company to make decisions that generate value for shareholders (Denis and McConnell, 2003). Nevertheless, the literature does not reach a consensus regarding the most appropriate CG mechanisms. Based on prior studies and intending to capture the main aspects of CG, we consider the board size, the shareholder voting rights and the ATPs.

Taking into account all the above, our study offers important contributions to the literature. First, we provide evidence that acquirers realize significant positive excess returns in short event windows around the announcement date. Our results are in contrast with Mateev and Andonov (2016) who focus on an overlapping sample period (2003-2010) as well as with Campa and Hernando (2004) and Martynova and Renneboog (2006, 2011) who examine European M&As before 2001. Second, we find board size, voting rights and ATPs to be significant variables of value creation for European acquirers. In particular, boards with more than 8 directors have a significant negative effect on the abnormal returns for acquiring firms, suggesting that larger boards are more ineffective than smaller ones due to inefficient coordination and communication. (Al-Bassam *et al.*, 2015; Jensen, 1993; Nerantzidis and Tsamis, 2017; Samaha *et al.*, 2012). Similar conclusions can be drawn for the other two examined measures, namely, voting rights and corporate control (in the form of ATPs). Our results suggest that higher voting rights generate higher gains for acquirers, as increased shareholder voting rights allow shareholders to monitor the decision-making of managers more closely and to have a decisive role in major corporate decisions (Gompers *et al.*, 2003). As far as ATPs are concerned, empirical evidence shows that firms with increased numbers of anti-takeover measures generate higher excess returns to their shareholders. This finding contradicts Masulis *et al.* (2007) who find a negative correlation between ATPs and announcement-period abnormal returns for the US firms and Bauguess and Stegemoller (2008) who find no systematic relation between G index and bidder returns. This may be attributable to the more active role of the market for corporate control in the USA – contrary to Europe – and the differences in CG framework (Bauer *et al.*, 2004; Sergakis, 2015) and legal systems (La Porta *et al.*, 2002). All in all, the discussion of the above issues highlights the need for more empirical consideration of the effect of CG mechanisms that may have an influence on M&A value creation. This type of research could assist market participants and scholars to better understand how and through which channels CG shapes a major strategic decision, namely, M&As (Aktas *et al.*, 2016).

The rest of this paper is organized as follows: Section 2 reviews the related literature and addresses the challenges and opportunities in CGSs. Section 3 describes the data and presents the methodology used in our analysis. Section 4 presents the empirical findings

and Section 5 discusses the conclusions of the paper and suggests policy implications, as well as areas for further research.

2. Related literature and challenges in corporate governance statements

Our study relates to at least two strands of literature literatures (i.e. M&As and CG). The first is research on the economic impact of M&As on the shareholders of acquiring firms. The second is a literature focused explicitly on the effects of CG mechanisms on the firm's effectiveness. In this section, we review the related prior literature and describe how our study could contribute to the challenges of CGSs.

2.1 Prior empirical studies

The wealth effects of M&As on acquiring firms attracted particular research interest over the past decades. However, the empirical findings are mixed and do not allow the formation of generalized conclusions. The bulk of research work is focused on the US market where the recent study of [Alexandridis et al. \(2017\)](#) finds that M&As generate more value for acquirers after 2009 than ever before due to the quality-enhancing role of CG. Prior studies reveal mainly negative or at best insignificant results for acquirers ([Alexandridis et al., 2013](#); [Moeller et al., 2005](#); [Fuller et al., 2002](#); [Andrade et al., 2001](#); [Servaes, 1991](#); [Bradley et al., 1988](#); [Travlos, 1987](#)), while only few of them suggest that M&As are value-enhancing strategic decisions ([Francis et al., 2008](#); [Loderer and Martin, 1990](#); [Asquith et al., 1983](#)). As far as European acquirers are concerned, the majority of studies ([Mateev and Andonov, 2016](#); [Campa and Hernando, 2004](#); [Martynova and Renneboog, 2006, 2011](#)) reveals insignificant abnormal returns.

The analysis of CG mechanisms (such as board size, ATPs and voting rights) as significant determinants of corporate success constitutes a favorable research area, as well. Prior empirical studies indicate that board size plays a significant role in board effectiveness. For example, [Khorana et al. \(2007\)](#) argue that smaller boards are more effective than oversized ones. This implies that in smaller boards directors can coordinate and communicate more effectively ([Al-Bassam et al., 2015](#); [Jensen, 1993](#); [Nerantzidis and Tsamis, 2017](#); [Samaha et al., 2012](#)). Along the same lines, there is a considerable number of studies that indicate a negative relationship between board size and firm value ([Conyon and Peck, 1998](#); [De Andres et al., 2005](#); [Jensen, 1993](#); [Shivdasani and Yermack, 1999](#)), while others fail to find a significant relationship ([Boone et al., 2007](#); [Lehn et al., 2009](#)). Another branch of CG literature is focused on the effect of ATPs on acquirers' gains and has produced inconclusive findings. For instance, [Bebchuk et al. \(2009\)](#) and [Gompers et al. \(2003\)](#) argue that ATPs can reduce firm value. Similarly, [Masulis et al. \(2007\)](#) find a negative relation between ATPs and announcement-period abnormal returns for the US firms. However, [Bauguess and Stegemoller \(2008\)](#) find no systematic relation between G index and bidder returns.

Finally, there is a significant body of theoretical and empirical literature in CG that investigates the legal protection of minority investors. Shareholder voting rights determine the level of legal protection and serve as a means for solving agency problems ([Gompers et al., 2003](#)). Specifically, [Bainbridge \(2005\)](#) suggests that voting is effectively an exercise of managerial power, as both strategic and tactical business decisions can be made efficiently through voting. For instance, [Bhagat and Bolton \(2008\)](#) mention that managers could increase leverage beyond "optimal capital structure" to increase the voting power of their equity stakes and reduce the likelihood of a takeover. In general, voting power may be used either by insiders to control the firm without holding large equity stakes or by outsiders if the super-voting shares are traded ([Li et al., 2008](#)). In the case of outsiders, increased shareholder voting rights may allow shareholders to monitor the decision-making of

managers more closely and to have a decisive role in major corporate decisions, such as M&As.

2.2 Challenges and opportunities in corporate governance statements

It is generally accepted that the most important EU initiative on CG regimes was the adoption of the “Comply or Explain” approach through Directive 2006/46/EC. According to this Directive, companies whose securities are traded on stock exchanges must include CGSs in their annual reports (Sergakis, 2015). The statements are constructed based on the requirements of the code that each company decides to follow, containing all the related information. In CGSs companies are obliged to disclose both the compliance and noncompliance (i.e. explain the reasons for non-compliance) with all best practices (namely provisions) relative to the adopted CG. However, in practice, as Parum (2005, p. 704) observes, it is questionable whether:

[...] shareholders or stakeholders, after reading the CGSs in the annual reports, know who sits on the board, whether they are independent, what their qualifications are to handle the strategic challenges of globalization, and in so doing, strengthen companies' competitiveness [...].

To put it differently, do market participants find information that they can use to support their decision-making?

According to theory, companies have many more incentives to disclose information beyond hard and/or soft laws (codes of good governance) to elevate transparency and increase shareholders' value. This illustrates that CGSs should act as a communication channel between the company and its shareholders and stakeholders, as well (Amran *et al.*, 2010). Nevertheless, in our study we try to identify and capture information (i.e. in CGSs and annual reports) beyond “conventional best practices” such as ATPs and voting rights. Our research suggests that this “information”, when presented, is qualitative without being easily quantifiable. Thus, the complexity of identifying legal provisions and codifying them into quantifiable variables, such as the La Porta *et al.* (2002) six-component anti-director index on investor protection, hinders the decision-making process. Finally, our findings suggest that despite the continuous convergence efforts toward an integrated European Union, the CG landscape seems to be segmented due to different national regulatory frameworks and financial structure across the member countries (Nenova, 2006). Thus, we suggest that European countries try to overcome any cultural, legislative and structural differences (Chiu, 2015) to create a unified and harmonized CG regime.

3. Data and methodology

3.1 Sample

The sample is comprised of M&A deals announced by European firms between 01/01/2003 and 31/12/2017. Datastream recorded 160,613 such deals during the sample period. For a deal to remain in the sample, it must meet the following criteria:

- The acquirer and the target are listed firms and their headquarters are located in EU, Switzerland and Norway.
- The transaction is completed before the end of the sample period.
- The acquirer has financial statement information at the year-end prior to the announcement available from Worldscope and stock price data for 270 days before the announcement date and 20 days after the announcement date in DataStream.
- The acquirer is included in the ESG databases provided by Thomson Reuters[1].

We omit buybacks, exchange offers and recapitalizations; exclude all bidder and target firms from the financial industry (SIC codes 6000-6999) because of the highly regulated

nature of the industry. To avoid the effects of very small transactions, we filter out those M&As with deal values less than \$1m and deal ratios (calculated as the ratio of the deal value to the acquirer market capitalization measured on the 21st trading day before the announcement day) below 1 per cent. To avoid the confounding effects of multiple bids, we exclude the deals from the same acquirers within 20 days (Fuller *et al.*, 2002). This selection process results in 349 deals that satisfy the criteria and remain in the sample.

3.2 Sample summary statistics

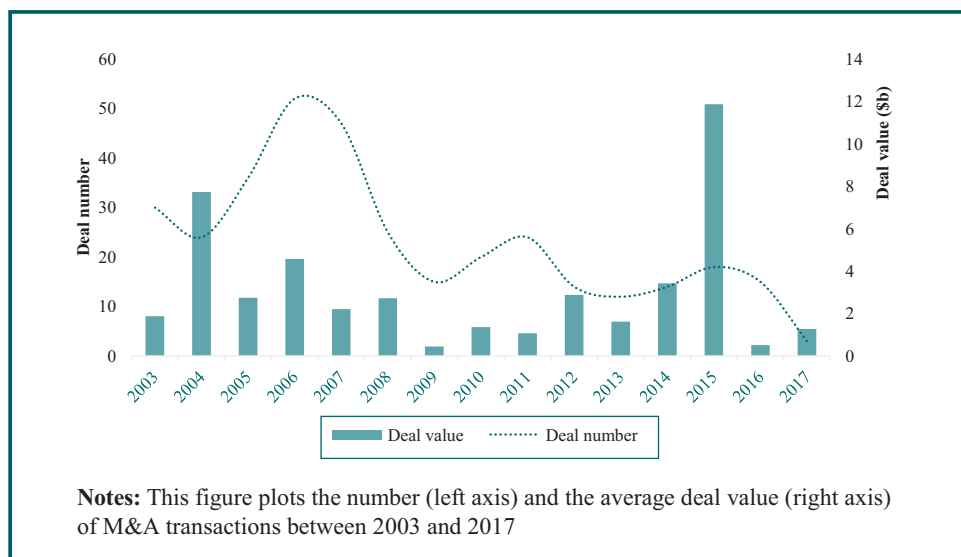
Table I and Figure 1 show the distribution of M&A deals over time. The starting point of the sample coincides with the commencement of the sixth merger wave (Alexandridis *et al.*, 2012) which ended in 2007 due to the outbreak of the global financial crisis. M&A deals started to recover during the first years of the current decade and the rebound was more pronounced in terms of dollar value. The average deal value skyrocketed in 2015 and it fell to previous levels over the following two years. From 2010 onward, firms opted to expand their activities geographically, thus increasing the share of cross-border transactions. On the contrary, the percentage of diversifying deals decreased after 2010 indicating a tendency of firms to focus more on the existing business sectors. Overall, the majority of M&A deals are domestic (56 per cent) and diversifying (68 per cent).

Table II depicts the geographic dispersion of acquirers and targets during the sample period. The dominant market for corporate control in Europe is the UK considering that it is the country with the largest number of M&A deals. In particular, UK firms are acquirers in 81 deals (23 per cent of the sample) and targets in 89 deals (26 per cent of the sample). In continental Europe, the most active market for corporate control is France, followed by Germany and Spain. Firms from these countries are acquirers in 46, 37 and 31 deals, respectively, representing 33 per cent of the sample. Similarly, 48 targets are from France and 29 targets are from Germany and Spain, representing 30 per cent of the sample. A market that should not be underestimated is the Scandinavian, since more than 10 per cent of bidders and targets are from Norway and Sweden. In contrast, the activity of M&As in

Table I Sample descriptive statistics

Year (A)	M&As (B)	Mean value (C)	Domestic (D)	Cross-border (E)	Horizontal (F)	Diversifying (G)
2003	30	1880.03	17	13	14	16
2004	24	7690.97	17	7	6	18
2005	36	2602.71	16	20	5	31
2006	52	4453.50	31	21	18	34
2007	47	2143.68	27	20	18	29
2008	25	1557.78	12	13	9	16
2009	15	411.46	10	5	3	12
2010	20	1297.65	12	8	4	16
2011	24	987.81	14	10	8	16
2012	14	2866.93	8	6	3	11
2013	12	1623.24	7	5	3	9
2014	14	3430.83	8	6	6	8
2015	18	11861.74	9	9	6	12
2016	15	456.33	8	7	7	8
2017	3	1274.31	1	2	2	1
Total	349	2969.26	197 (56%)	152 (44%)	112 (32%)	237 (68%)

Notes: This table describes the sample of 349 deals announced and completed among the 28 European member countries (Plus Norway and Switzerland) between 2003 and 2017. Column B reports the distribution of deals across announcement years. Column C reports the average deal value (\$m) across announcement years. Columns D and E report the distribution of domestic and cross-border deals across announcement years, respectively. The final two columns report the distribution of horizontal and diversifying deals across announcement years

Figure 1 Annual distribution of deals and deal value

many countries of Central Europe is remarkably low, as it is in countries that joined the European Union in 2004 and 2007.

3.3 Measurement of abnormal returns

We build on the classic event study methodology to measure the wealth effects deriving from merger announcements on the stock price of European acquirers. We calculate mean and median cumulative abnormal returns (CARs) with the ordinary least squares estimation of the Carhart four-factor model (Carhart, 1997), which expands on the classic three-factor model of Fama and French (1993) by introducing an additional stock momentum factor. We use European Fama-French and momentum factors, which we obtain from the public data library^[2] of Kenneth R. French.

The four-factor model is used as shown in equation (1):

$$\hat{R}_{it} - R_{ft} = \alpha_i + \beta_{MKT}(R_{MKT} - R_{ft}) + \beta_{SMB}SMB + \beta_{HML}HML + \beta_{MOM}MOM + \varepsilon_{it} \quad (1)$$

where:

$\hat{R}_{it}$ is the expected return of company i at day t

R_{ft} is the risk-free return

α_i is the model's intercept

β_{MKT} , β_{SMB} , β_{HML} , β_{MOM} are the factors' coefficients

SMB is a size factor, capturing the performance of low capitalization firms over high capitalization firms

HML is a value factor, capturing the excess returns of firms with high book-to-market value over firms with low book-to-market-value

MOM is a momentum factor capturing the excess returns of past good performing stocks over poor performing stocks.

The estimation period of model coefficients is 250 days preceding the first day of the event window $(-270, -21)$ while the abnormal returns are aggregated over a 41-day event

Table II																							Geographic scatter of M&A transactions																						
Acquirer country		AU	BE	HR	CZ	DE	FI	FR	GE	GR	HU	Target country							IT	LT	LU	NL	NO	PL	PT	IR	RO	ES	SE	CH	UK	Total													
AU	1							2	2		1															1						7													
BE		2						1	2																						1	6													
DE						3		1																							5														
FI							3	2																							1	6													
FR		2			1			28	1	1																					3	46													
GE	1							2	11	3																					5	37													
GR										7																						7													
IT		1						2	3	2																					1	21													
LU																															1														
NL		1	1					2	1																	1					4	19													
NO							1																									22													
PL								1																								7													
PT																																6													
IR																																2													
ES								1	3																						1	3													
SE							2																								3	31													
CH							1	3	4	1	1																				4	2	16												
UK		1	1				1	3	2																						1	11	4	28											
Total	2	7	2	2	3	8	48	29	14	2	15	1	2	18	25	7	7	1	2	29	17	19	89	349																					
Notes: AU = Austria; BE = Belgium; HR = Croatia; CZ = Czech Republic; DE = Denmark; FI = Finland; FR = France; GE = Germany; GR = Greece; HU = Hungary; IT = Italy; LT = Lithuania; LU = Luxembourg; NL = Netherlands; NO = Norway; PL = Poland; PT = Portugal; IR = Republic of Ireland; RO = Romania; ES = Spain; SE = Sweden; CH = Switzerland; UK = United Kingdom																																													

window $(-20, 20)$. The 41-day window serves a threefold purpose of capturing any potential information leakage prior to the event announcement, allowing the examination of value creation beyond the initial information shock and providing comparable results to other similar studies. Moreover, we use multiple windows to obtain some insight into the timeframe in which excess value is generated. We apply eight different event windows: two pre-announcement windows $(-20, 0)$ and $(-5, 0)$; two short-run windows around the announcement day $(-1, 1)$ and $(0, 1)$; two post-announcement windows $(0, 20)$ and $(0, 5)$; and two symmetric windows surrounding the announcement day $(-5, 5)$ and $(-20, 20)$.

The abnormal returns are calculated as shown in [equation \(2\)](#):

$$AR_{it} = R_{it} - \hat{R}_{it} \quad (2)$$

where:

AR_{it} is the abnormal return of company i at day t

R_{it} is the realized return of share i at day t

$\hat{R}_{it}$ is the expected return of company i at day t from [equation \(1\)](#)

CARs are calculated as shown in [equation \(3\)](#):

$$CAR_{i(t1,t2)} = \sum_{t=t_1}^{t_2} AR_{it} \quad (3)$$

We assess the statistical significance of our results with the Patell Z test ([Patell, 1976](#)) and the Rank test ([Corrado, 1989](#)). Each abnormal return is standardized with its estimated standard deviation. The Patell test is used as shown in [equation \(4\)](#):

$$T_{Patell} = \frac{1}{\sqrt{N}} \sum_{i=1}^N \frac{SCAR_{i(t1,t2)}}{S(SCAR_i)} \quad (4)$$

where:

$SCAR_{i(t1,t2)}$ is the standardized cumulative abnormal return of firm i

$S(SCAR_i)$ is the standard deviation of the standardized CARs of firm i

The Corrado test is used as shown in [equation \(5\)](#).

$$T_{Corrado} = \sqrt{L_2} \left(\frac{\bar{K}_{T_1, T_2} - 0.5}{SK} \right) \quad (5)$$

where:

$\bar{K}_{T_1, T_2}$ is the mean rank across firms and time in the event window

$S_{\bar{K}}$ is the standard deviation defined as:

$$S_{\bar{K}} = \sqrt{\frac{1}{L_1 + L_2} \sum_{t=T_0}^{T_2} \frac{N_t}{N} (\bar{K}_t - 0.5)^2} \quad (6)$$

L_2 is the event window length, L_1 is the estimation window length, T_0 is the first day of the estimation window and T_2 is the latest day of the event window

N_t is the number of non-missing returns across firms.

3.4 Multivariate analysis

We intend to explain acquirer's gains using a number of factors deriving from M&A literature. To assess the effect of CG on bidders' excess returns, after controlling for the

effects of other factors, we estimate a multi-variable linear regression model as shown in equation (7):

$$CAR_{i(t1,t2)} = \beta_0 + \sum_{i=1}^N \beta_i X_i + \varepsilon_i \quad (7)$$

where:

β_0 is the model's intercept

$\beta_i X_i$ is a matrix of independent variables

ε_i is the error term.

In equation (7), the dependent variable (CARs) is the cumulative excess returns of bidders around the announcement day as estimated in equation (3). Our main interest lies in the variables representing the framework of CG. The measures of CG that represent the variables of interest are:

- the board size (*BSIZE*);
- the shareholder voting rights (*VRS*); and
- the anti-takeover provisions (*ATP*).

Board size is a dummy variable that takes the value of 1 if the number of board of directors is greater than 7 and 0 if it is up to 7 (i.e. for the construction of the variable we used information provided by CGSs and/or annual reports, while we verified this information with ESG databases – Thomson Reuters). For the other two measures we use the Percentile Rank Score for each company which is based on:

- the number of companies that are worse than the one under examination;
- the number of companies having the same value; and
- the number of companies having a value at all.

The percentile rank score is calculated as shown in equation (8):

$$Score = \frac{n. \text{ of companies with a worst value} + \frac{n. \text{ of companies with the same value}}{2}}{n. \text{ of companies with a value}} \quad (8)$$

Based on merger literature, we use a set of firm-specific control variables to capture any possible effects on bidders' value creation. Zhang (2006) asserts that in the case of firms with long history more information is available to the market, which in turn mitigates information asymmetry between the firm and investors. Draper and Paudyal (2006) highlight asymmetric information as a significant determinant of bidder returns. Therefore, we use the age of the acquirer (*AGE*), measured by the natural logarithm of the number of days between the acquisition announcement date and the first date on which Datastream has data for the company, to account for possible effects of information asymmetry. Prior literature suggests growth opportunities as a significant determinant of abnormal returns for acquiring firms. Rau and Vermaelen (1998) argue that value acquirers (firms with a low market-to-book value ratio) outperform glamour acquirers (firms with a high market-to-book value ratio) in the USA. Sudarsanam and Mahate (2003) and Andriosopoulos *et al.* (2016) reach similar conclusions for the UK firms. However, Conn *et al.* (2005) reveal that glamour acquirers perform poorly only in case they acquire public targets, as opposed to private ones. Besides, Alexandridis *et al.* (2008) report a statistically insignificant relation between the abnormal returns of acquirers and their market-to-book ratio. Therefore, we use the ratio of market-to-book value of equity (*MKTB*) 21 days preceding the announcement date, to control for the effect of growth opportunities to acquirers. Lang *et al.* (1989) and Servaes (1991) show that bidders with high Tobin's *q* ratio experience significant positive returns when they engage in mergers and tender offers, while bidders with low *q* ratio have

significant negative abnormal returns. However, [Dong et al. \(2006\)](#) fail to confirm the results of past research, while [Moeller et al. \(2005\)](#) find an economically important negative relation between Tobin's q and bidder returns. Therefore, we use the Tobin's q ratio (*TOBINQ*) measured as the ratio of the firm's market value to the replacement value of assets, to control for possible implications of bidders' misvaluation. Deriving from [Jensen's \(1988\)](#) free cash flow theory, we also use bidders' leverage. A high leverage level reduces future cash flows, limiting managerial discretion. Besides, managers have an incentive to improve the performance of their firms to meet future obligations, given that in the opposite case they may be replaced. [Maloney et al. \(1993\)](#), [Moeller et al. \(2005\)](#) and [Masulis et al. \(2007\)](#) find that bidders with higher leverage show higher abnormal returns. Hence, we use Leverage (*LEV*), measured as the ratio of the firm's total debt to its total assets at year-end prior to the merger announcement. [Fuller et al. \(2002\)](#) suggest that the relative size of the deal may affect acquirers' excess returns. Thus, we use the relative size of the deal (*RSIZE*), measured by the ratio of the deal value to acquirers' market value 21 days preceding the announcement date. Also, we use a profitability measure, the Return on Assets (*ROA*) at year-end prior to the merger announcement, as a proxy for the strength and efficiency of acquirers, as well as their ability to generate value for their shareholders.

Apart from firm-specific variables, we use the deal characteristics to account for the potential implications of the domiciles of the targets, the scope of the transaction, the deal attitude and value. Although geographical and product diversification likely affect the wealth effects of M&As, the literature fails to reach a consensus on whether it enhances or destroys a firm's value. Firms may benefit from cross-border deals due to imperfect markets that cause asset mispricing, differences in tax and regulatory regimes between the domestic and the foreign country and favorable exchange rate movements ([Goergen and Renneboog, 2004](#); [Moeller and Schlingemann, 2005](#)). However, firms involved in international transactions undertake higher business risk owing to information asymmetries and differences in culture, customer preferences, CG and regulation ([Erel et al., 2012](#)). Geographic-focused transactions provide the opportunity to exploit cost economies deriving from operational synergies, overlapping activities, shared technology and better market awareness. Therefore, we use a dummy variable (*GEO*) that is assigned the value of 1 for cross-border deals and 0 for domestic deals, to control for the potential effects of geographical diversification.

Firms can mitigate risks and thereby increase value through product diversification. Diversified firms may exploit economies of scope if they possess excess capacity in resources and capabilities that can be transferred across different industries ([Martin and Sayrak, 2003](#)). Corporate diversification allows firms to reduce earnings volatility, which in turn lowers the reliance on external financing and the cost of capital. However, diversification can cause higher monitoring costs and firms may suffer from their inexperience in managing unrelated activities. Also, the diversity of activities exacerbates managerial agency problems resulting in a diversification discount ([Akbulut and Matsusaka, 2010](#); [Laeven and Levine, 2007](#)). [Erdorf et al. \(2013\)](#) review the related literature concluding that the effect of diversification on firm value is heterogeneous and is related to certain industry settings, economic conditions and governance structures. Therefore, we use a dummy variable (*IND*) that is assigned the value of 1 for focused deals (i.e. bidder and target share the same two-digit Standard Industrial Classification (SIC) code) and 0 for diversifying deals, to control for the potential effects of product diversification.

The literature highlights the status of the bid as an important determinant of value creation for bidders. In an active and competitive market for corporate control, hostile takeovers act as a disciplinary force to remove managers of poorly performing firms. The early studies of [Jensen \(1988\)](#), [Morck et al. \(1988\)](#) and [Martin and McConnell \(1991\)](#) find higher shareholder value creation from hostile deals compared to friendly ones in the US. [Sudarsanam and Mahate \(2006\)](#) reveal similar findings for the UK, while [Goergen and Renneboog \(2004\)](#) present opposing results for European firms. Therefore, we use a

dummy variable (*ATT*) that is assigned the value of 1 for hostile bids and 0 otherwise, to control for the potential effects of deal attitude.

Alexandridis *et al.* (2013), among others, suggest that acquisitions with large deal size destroy more value for acquiring companies. Large deals are associated with increased complexity which makes it more difficult for acquirers to exploit economic benefits. Therefore, we use the natural logarithm of the deal value (*DVAL*).

Finally, we consider economic uncertainty since it is commonly accepted that it affects the economy and harms the level of investment of firms (Abel, 1983; Bernanke, 1983; Bloom, 2009; Bloom *et al.*, 2007; Dixit and Pindyck, 1994; McDonald and Siegel, 1986). Baker *et al.* (2016) introduced an index of economic policy uncertainty (EPU) that serves as a proxy for the intensity of policy-related economic uncertainty in a country or a region. Bonaime *et al.* (2016) and Nguyen and Phan (2017) investigate the relationship between EPU and M&As in the USA from mid-1980s to 2014, confirming a significant link between policy uncertainty and the wealth effects of merger deals. Therefore, we construct the policy uncertainty variable (*EPU*), measured by the natural logarithm of the arithmetic mean of the European EPU index[3] monthly values of the fiscal year prior to the announcement date, to control for the potential effects of uncertainty (Table III).

Table III Summary statistics of regression variables

Variables	N	Mean	Median	SD	Q1	Q3
<i>Panel A. Corporate governance variables</i>						
Board size	349	0.834	1	–	–	–
ATPs score	349	0.566	0.647	0.223	0.576	0.668
Voting rights score	349	0.459	0.688	0.321	0.028	0.697
<i>Panel B. Firm-specific variables</i>						
Firm age	349	8.608	8.712	0.821	8.313	9.183
Return on assets	349	6.821	6.010	7.159	3.720	9.270
Market-to-book ratio	349	3.648	2.161	37.553	1.347	3.168
Tobin's <i>q</i> ratio	349	1.595	1.368	0.791	1.158	1.779
Leverage	349	27.635	26.010	15.634	15.700	36.960
Relative size	349	3.460	0.075	30.152	0.021	0.337
<i>Panel C. Deal-specific characteristics</i>						
Geographic focus	349	0.436	0	–	–	–
Product focus	349	0.682	1	–	–	–
Deal attitude	349	0.797	1	–	–	–
Deal value	349	6.303	6.231	1.856	5.029	7.429
<i>Panel D. Macroeconomic environment</i>						
Economic policy uncertainty	349	123.310	119.520	41.316	88.340	146.350

Notes: This table reports summary statistics on the explanatory/control variables. Board size (BSIZE) is a dummy variable that takes the value of 1 if the board size at end year prior to the merger announcement is greater than 7 and 0 otherwise. Anti-takeover provisions score (ATP) is the ASSET4 score of the anti-takeover devices in excess of two used by the firm. Voting rights score (VRS) is the ASSET4 score of the equality of a firm's shares voting rights. Firm age (AGE) is the natural logarithm of the days between the acquisition announcement date and the first date on which datastream has data for the company. Return on assets (ROA) is the return on assets at year-end prior to the merger announcement. Market-to-book ratio (MKTB) is the market-to-book value of the firm's equity 21 days preceding the announcement date. Tobin's *q* ratio (TOBINQ) is the ratio of the firm's market value to the replacement value of assets. Leverage (LEV) is the year-end ratio of the firm's total debt to its total assets. Relative size (RSIZE) is the ratio of the deal value to the firm's market value 21 days preceding the announcement date. Geographic focus (GEO) is a binary variable that takes the value of 1 if the deal is a cross-border one and 0 if its domestic. Product focus (IND) is a binary variable that takes the value of 1 if the deal is horizontal and 0 otherwise. Deal attitude (ATT) is a binary variable that takes the value of 1 if the deal is hostile and 0 otherwise. Deal value (DVAL) is the natural logarithm of the deal value (\$m). Economic policy uncertainty index (EPU) is the European monthly measurement of policy-related economic uncertainty

4. Empirical findings

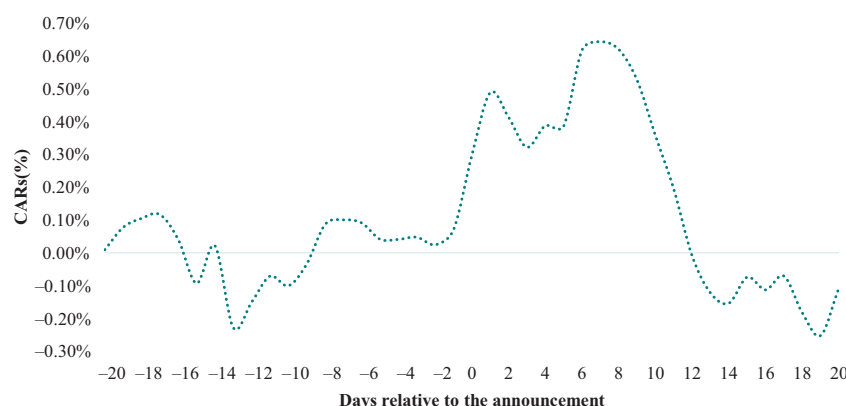
This section begins with a discussion of the results of the event study, followed by a discussion of the results of the cross-section regression analysis for the acquirers' excess returns, including implications of the CG measures.

4.1 Abnormal returns of acquirers

Figure 2 illustrates the CARs of European acquiring firms over the 41-day window (−20,20) surrounding the announcement date based on the four-factor model. The graph shows that, on average, acquirers enjoy a significant market reaction on the day of the announcement (day 0). The positive reaction commences on the announcement day and lasts for a few days thereafter. This is a short-lived reaction as there is a considerable downward trend a week after the announcement. The negative trend unfolds toward the end of the window, eliminating any value gains. Eventually, bidders' CARs rebound to the pre-announcement period level.

We assess the wealth effects of M&As announcements on acquiring firms in more detail in Table IV. Acquirers realize significant positive excess returns in short event windows around the announcement date. The three-day and two-day mean (median) CARs are 0.46 per cent (0.42 per cent) and 0.42 per cent (0.31 per cent), respectively, both statistically significant at the 1 per cent (5 per cent) level. Acquiring firms show very low abnormal returns during periods prior to and after the announcement date, which are insignificant. The share of acquirers with positive abnormal returns is fairly constant above 50 per cent in almost all the applied event windows. The results of this study contradict Mateev and Andonov (2016) who calculate insignificant CARs for bidders in short windows (from 3 to 16 days) around the announcement day during the period from 2003 to 2010. Also, our results diverge from Campa and Hernando (2004) and Martynova and Renneboog (2006, 2011) who focus on European M&As before 2001. The former examined M&A announcements over the period 1998-2000, while the latter analyze European acquisitions undertaken during the fifth

Figure 2 Cumulative abnormal returns for bidders



Notes: This figure plots the cumulative abnormal returns of a 41-day window for a sample of 349 deals announced and completed among the 28 European member countries (plus Norway and Switzerland) between 2003 and 2017. Abnormal returns are calculated with least squares regression using the four-factor model (Carhart, 1997) and the European Fama-French and momentum factors in an estimation period of 250 days preceding the first day of the event window (−270, −21)

Table IV Event study results

Event windows	Mean (%)	Median (%)	Positive CARs (%)	Patell	p-value	Corrado	p-value
<i>Pre-announcement</i>							
(−20,0)	0.30	0.24	49	0.53	0.60	0.88	0.38
(−5,0)	0.21	−0.29	53	1.02	0.31	0.87	0.38
<i>Announcement</i>							
(−1,1)	0.46***	0.42 ^{††}	57	3.50	0.00	2.22	0.03
(0,1)	0.42***	0.31 ^{††}	56	3.92	0.00	2.34	0.02
(−5,5)	0.30	−0.24	53	0.82	0.41	1.00	0.32
(−20,20)	−0.11	0.36 [†]	52	−0.57	0.57	0.07	0.95
<i>Post-announcement</i>							
(0,20)	−0.18	0.16	51	−0.64	0.53	−0.49	0.62
(0,5)	0.32	0.18	52	1.37	0.17	1.03	0.30

Notes: This table presents the results of the event study for a sample of 349 deals made among the 28 European member countries (Plus Norway and Switzerland) between 2003 and 2017. Abnormal returns are calculated with least squares regression using the four-factor model (Carhart, 1997) and the European Fama-French and Momentum factors in an estimation period of 250 days preceding the first day of the event window (−270, −21). The statistical significance of the results is assessed with the Patell Z test (Patell, 1976) and the Rank test (Corrado, 1989). *, **, *** denote the statistical significance with the parametric test at 10, 5 and 1 per cent level, respectively. †, ††, ††† denote the statistical significance with the non-parametric test at 10, 5 and 1 per cent level, respectively

takeover wave (1993-2001). Both studies find insignificant abnormal returns to acquiring firms except for the existence of a significant run-up starting at least 40 days prior to the announcement day. The results obtained by this study are close to the ones reported by Goergen and Renneboog (2004). The authors examine large acquisitions (deal value greater than \$100m) over the period 1993-2000 and report significant (at the 1 per cent level) abnormal returns of 0.7 per cent and 1.18 per cent in the (−1,0) and (−2,2) event windows, respectively, and insignificant returns in windows with longer duration.

4.2 Cross-sectional analysis of acquirers' gains

The following paragraphs present the results of univariate and multivariate cross-sectional regression analyses where the dependent variable is the bidders' CAR of the (0, 1) event window. We apply the 2-day CAR because of its statistical significance and because it refers to a small period around the announcement. To increase the robustness of results we:

- winsorize at the 1 and 99 per cent levels to reduce the effect of possibly spurious outliers;
- quantify the severity of multicollinearity using the variance inflation factors (VIF); and
- use robust standard errors.

Table V documents the estimates of multivariate analysis [equation (7)] that account for the effects of CG measures in conjunction with several factors that are likely to affect the excess returns of bidders.

Initially, we estimate univariate models to identify the potential effect of each variable in isolation. The results indicate that the firm-specific variables (models 1-6), apart from firm age and leverage, have a positive effect on bidders' CARs. However, only the return on assets is statistically significant (at the 1 per cent level). The economic significance is particularly large as a one standard deviation increase in return on assets is associated with an increase in CARs of 67 basis points. Our implication is that the market assumes that bidders with high profitability are in a better position to exploit synergies and create value through mergers. The deal-specific variables (models 7-10) appear to have a positive effect

Table V The determinants of bidders' CARs: univariate and multivariate analyses

Models	C	AGE	ROA	MKTB	TOBINQ	LEV	RSIZE	GEO	IND
<i>Panel A. Univariate analysis</i>									
1	1.900	-0.172 (-0.557)							
2	-0.220		0.094*** (2.685)						
3	0.364			0.020 (0.187)					
4	0.081				0.213 (0.641)				
5	0.509					-0.003 (-0.214)			
6	0.375						0.028 (1.073)		
7	0.362							0.128 (0.264)	
8	0.389								0.090 (0.174)
9	0.417*								
10	0.227								
11	1.234								
12	1.711***								
13	-0.801								
14	-0.357								
<i>Panel B. Multivariate analysis</i>									
15	0.946		0.082** (2.327)						
16	1.939	-0.043 (-0.139)	0.116*** (2.670)	-0.005 (-0.034)	-0.564 (-1.123)	0.004 (0.250)	0.025 (0.961)	0.173 (0.345)	0.162 (0.302)
17	1.228								
18	-1.363**		0.092*** (2.662)						
19	0.243	-0.132 (-0.483)	0.118*** (2.731)	-0.001 (-0.010)	-0.454 (-0.903)	0.003 (0.214)	0.022 (0.830)		
20	-1.852							0.263 (0.518)	0.114 (0.212)
21	-0.883*		0.088** (2.517)	-0.007 (-0.047)	-0.509 (-1.014)	0.003 (0.167)	0.023 (0.888)		
22	0.392	-0.090 (-0.296)	0.118 (2.738)***						
23	-0.885							0.247 (0.488)	0.116 (0.216)
24	2.257	-0.032 (-0.104)	0.110** (2.482)	-0.004 (-0.031)	-0.584 (-1.146)	0.003 (0.168)	0.026 (0.988)	0.179 (0.355)	0.156 (0.290)
25	0.159	-0.122 (-0.400)	0.111** (2.498)	0.001 (0.010)	-0.475 (-0.934)	0.002 (0.108)	0.022 (0.820)	0.257 (0.506)	0.144 (0.267)
26	0.551	-0.91 (-0.296)	0.115** (2.597)	-0.005 (-0.034)	-0.522 (-1.027)	0.002 (0.112)	0.024 (0.924)	0.241 (0.477)	0.135 (0.250)

Notes: This table reports least squares regressions of the abnormal returns in the event window (0,1) and a set of control variables. The sample consists of 349 M&A deals completed by 28 European member countries (plus Norway and Switzerland) between 2003 and 2017. Abnormal returns are calculated using the four-factor model (Carhart, 1997) and the European Fama-French and momentum factors in an estimation period of 250 days preceding the first day of the event window (-270, -21). Control variables are winsorized at the 1 and 99% level. For a detailed description of the variables see Table A1 in the Appendix. *, **, *** denote the statistical significance at the 10, 5 and 1 per cent level, respectively (continued)

Table V

Models	ATT	DVAL	EPU	BZIZE	ATP	VRS	R ²	F	VIF
<i>Panel A. Univariate analysis</i>									
1							0.001	0.333	1.000
2							0.020	7.208***	1.000
3							0.000	0.035	1.000
4							0.001	0.411	1.000
5							0.011	0.046	1.000
6							0.003	1.152	1.000
7							0.000	0.070	1.000
8							0.000	0.030	1.000
9	0.078(0.042)						0.000	0.002	1.000
10		0.030(0.230)					0.000	0.053	1.000
11			-0.007(-1.095)				0.003	1.198	1.000
12				-1.551**(-2.416)			0.017	5.839**	1.000
13					2.149**(2.002)		0.011	4.009**	1.000
14						1.690**(2.262)	0.015	5.116*	1.000
<i>Panel B. Multivariate analysis</i>									
15				-1.302**(-2.014)			0.032	5.665***	1.028
16				-1.414**(-2.120)			0.040	2.025*	1.405
17	-0.004(-0.002)	0.067(0.505)		-1.634**(-2.497)			0.018	1.272	1.044
18					2.045*(1.920)		0.031	5.476***	1.001
19					1.954*(1.813)		0.037	1.848*	1.390
20	-0.036(-0.019)	0.109(0.794)			2.520**(2.215)		0.014	1.007	1.077
21						1.534** (2.063)	0.032	5.764***	1.007
22						1.552** (2.056)	0.039	1.985*	1.392
23	0.181(0.098)	0.053(0.399)				1.789** (2.347)	0.016	1.127	1.044
24	-0.080(-0.043)	0.021(0.156)	-0.004(-0.653)				0.042	1.231	1.279
25	-0.204(-0.109)	0.060(0.424)	-0.005(-0.779)	-1.481**(-2.175)	2.309**(2.007)		0.040	1.171	1.292
26	0.022(0.012)	0.009(0.066)	-0.003(-0.401)			1.597** (2.063)	0.041	1.190	1.298

on the CARs; however, none of these are statistically significant. Similarly, the Economic Policy Uncertainty (model 11) has an insignificant effect on bidders' CARs.

Univariate analysis further shows that the CG measures also affect the gains of bidders. Board size (model 12) presents a negative coefficient, which is statistically significant at the 5 per cent level. In line with the literature, our results corroborate the argument that larger boards fail to take value-enhancing strategic decisions for their companies. It seems that larger boards suffer from poor communication and decision-making that undermine their effectiveness (Guest, 2009). Our results are similar with the findings of prominent scholars in the field who suggested that board size should be no greater than eight or nine for all firms (Jensen, 1993; Lipton and Lorsch, 1992). Board size continues being statistically significant at the 5 per cent level in all the multivariate models (models 15-17 & 24), thus it is counted as a significant variable of bidders' excess returns.

ATPs show a positive and statistically significant at the 5 per cent level variable both in univariate (model 13) and multivariate (models 18-20 and 25) models. The coefficient of the variable implies that as firms increase the number of anti-takeover measures, they generate higher wealth effects for their shareholders. To assess the economic significance of the coefficient, a one standard deviation in ATPs score is associated with an increase in CARs of slightly less than half a percentage point. This finding contradicts Masulis *et al.* (2007) who find a negative relation between ATPs and announcement-period abnormal returns for the US firms and Bauguess and Stegemoller (2008) who find no systematic relation. As far as Shareholder Voting Rights variable is concerned both univariate (model 14) and multivariate (models 21-23 & 26) analyses suggest that higher scores generate higher gains to acquirers. The coefficient is positive and statistically significant at the 5 per cent level across all model specifications. The economic significance of that variable is particularly large as a one standard deviation increase in shareholder voting rights score increases returns by more than half a percentage point. Increased shareholder voting rights allow shareholders to monitor more closely the decision-making of managers and to have a decisive role in major corporate decisions, such as M&As. Shareholders can deter management from engaging in value-destroying mergers driven by agency motives and/or hubris. Therefore, voting rights can serve as an effective CG mechanism, mitigating conflicts of interest arising due to agency problems.

5. Conclusion and recommendations

The corporate finance literature fails to reach any sort of consensus on the economic impact of M&As. From that perspective, one of the purposes of this study is to assess the effect of M&A announcements by European acquiring firms on their stock price. Our results indicate that acquirers realize significant positive excess returns in short event windows surrounding the announcement day. In addition, we find that these returns are affected by certain CG mechanisms such as board size, voting rights and ATPs. More specifically, boards consisting of 8 directors and more are negatively related with the acquirers' abnormal returns. Our evidence implies that smaller boards are more effective than oversized ones, which means that they communicate better and are more efficiently coordinated. Moreover, we show that higher voting rights generate higher gains for the acquirers, given that increased shareholder voting rights enable shareholders to play a more active role in the managerial decision-making, supporting value-enhancing strategic decisions. In the same direction, we suggest that ATPs are positively related with the abnormal returns of bidders, considering that they allow firms to protect their autonomy and to have sufficient time to make rational investment decisions. The different effect of ATPs on European firms relative to their US counterparts may be explained by the different legislative framework and the level of competition of the market for corporate control between the USA and Europe.

In terms of policy direction, our findings assist market participants in perceiving the efficacy of CG practices. Investors should realize the importance of certain CG measures (i.e. board

size, voting rights and ATPs) when investing in firms which play an active role in the market for corporate control. Firms may reconsider the effectiveness of boards with a large number of directors; it seems that monitoring and communication problems outweigh the benefits of diverse and multidimensional boards. Also, firms should be more cautious regarding the issue of dual class shares or shares with multiple voting rights, since the decision-making process becomes more effective as more voting power is given to all shareholders. Firms may adopt more ATPs; however, the large number and variety of such measures make the selection process difficult. Considering the effect of CG on wealth creation, regulatory authorities and supervisory agencies in Europe should formulate codes that will further enhance the effectiveness of the CG framework.

Nevertheless, this study is subject to caveats worth noting. We investigate the effect of CG on bidder returns in case they acquire listed European targets. Widening the sample, including private targets located within or outside Europe, could lead to different conclusions. Also, our study shows that CG has a significant effect on acquiring firms. Investigating the effect of CG on target firms will be important. Finally, we reach our conclusions using certain CG measures. It would be challenging to assess the effect of other CG mechanisms on the economic impact of M&As.

In this context and beyond addressing the above limitations, the findings of our study could raise new issues for further research. Scholars in the field could focus on the effect of CG on the wealth effects of M&As considering the different systems of CG. Also, they may address the need for specific policy actions and reforms that will enhance the CG framework toward value-enhancing strategic business decisions. It is, therefore, important for researchers to consider the interdisciplinary and multidimensional nature of CG and recognize the need for empirical work that will shed light on new perspectives.

Notes

1. Thomson Reuters compiles information for a company's relative ESG performance, commitment and effectiveness across 10 main themes (resource use, emissions, environmental product innovation, workforce, human rights, community, product responsibility, management, shareholders, CSR strategy) based on company-reported data. ESG databases cover over 6,500 public companies globally, across more than 400 different ESG metrics, from 2002.
2. Available at: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
3. To measure European policy-related economic uncertainty, Baker *et al.* (2016) construct an index based on newspaper articles regarding policy uncertainty. They draw on two newspapers per country, which are *Le Monde* and *Le Figaro* for France, *Handelsblatt* and *Frankfurter Allgemeine Zeitung* for Germany, *Corriere Della Sera* and *La Repubblica* for Italy, *El Mundo* and *El Pais* for Spain and *The Times of London* and *Financial Times* for the UK. They count the number of newspaper articles containing the terms "uncertain" or "uncertainty", "economic" or "economy" and one or more policy-related terms, and then, they scale the raw EPU count by a measure of the number of articles in the same newspaper and month. They standardize each newspaper-level monthly series to unit standard deviation prior to 2011 and average across newspapers by month to obtain the European EPU index, which is normalized to a mean of 100 prior to 2011. The European EPU index derives from an equal average across all 10 European newspapers.

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Appendix

Table A1 Description of variables

<i>Variables</i>	<i>Description</i>
BSIZE	BSIZE is a dummy variable that takes the value of 1 if the board size at end year prior to the merger announcement is greater than 7 and 0 otherwise. (Own calculations, ASSET4 – Datastream code: CGBSDP060)
ATP	ATP is the ASSET4 score of the anti-takeover devices in excess of two used by the firm. (ASSET4 – Datastream code: CGSRO06S)
VRS	VRS is the ASSET4 score of the equality of a firm's shares voting rights. (ASSET4 – Datastream code: CGSRO02V)
AGE	AGE is the natural logarithm of the days between the acquisition announcement date and the first date of the company's first record in Datastream. (Datastream code: BDATE)
ROA	ROA is the returns on assets ratio of the firm at year-end prior to the merger announcement. ROA is an efficiency proxy that measure the firm's ability to use its assets to generate profits. (Worldscope – Code: WC08326)
MKTB	MKTB is the market-to-book (also known as price-to-book) ratio at day 21 relative to the merger announcement. MKTB reflects the market value that is attached to a company's assets relative to their book value. (Datastream code: PTBV)
TOBINQ	Tobin's Q ratio of the firm at year-end prior to the merger announcement. Tobin's q is the market value of a company's assets divided by the replacement cost of the company's assets
LEV	LEV is the ratio of total debt to total assets of the firm at the year-end prior to the merger announcement. (Worldscope – Code: WC08236)
RSIZE	RSIZE is the ratio of the deal value to the firm's market value at day 21 relative to the merger announcement. (Own calculations – market value Datastream Code: MV)
GEO	GEO is a binary variable that accounts for the geographic scope of the deal. It takes the value of 1 if the deal is a cross-border one and 0 otherwise. (Datastream)
IND	IND is a binary variable that accounts for product scope of the deal. It takes the value of 1 if the deal is horizontal and 0 otherwise. (Datastream)
ATT	ATT is a binary variable that accounts for the deal attitude. It takes the value of 1 if the deal is hostile and 0 otherwise. (Datastream)
DVAL	DVAL is the natural logarithm of the deal value expressed in \$. (Datastream)
EPU	EPU is the value of the European economic policy uncertainty index at the month of the merger announcement. (Index available at www.policyuncertainty.com)

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